COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/08/2026	**DATE SUBMITTED:**	07/10/2026
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ACTIVITY NO. 1.1 Algorithm, Flowchart, Pseudocode

1. PROCEDURE OUTPUT

1. Compute the total and average of 3 quizzes then display the total and average.

1. Display program title
2. Prompt the user to input their name.
3. Do the following 3 times:
    a. Prompt the user to input the number of times and score of a quiz.
    b. Add the scores to the total
    c. Compute the average of every quiz
4. Display the results, name, and scores.
5. Do the following 3 times.
    a. Display the score and number of times
    b. Compute the grade of the quiz.
    c. Display PASSED if the grade is passing, or FAILED if the grade is failing.
    d. Display the grade of the quiz
6. Display the total and average score of the 3 quizzes.
7. Prompt the user to repeat the program. If not, the program ends.

Figure 1. Algorithm of the Solution to Problem 1.

```
START
  DISPLAY program title
  INITIALIZE quizScores[3], quizItems[3], quizTotal, quizAvg, name, yesNo

  DO
    SET quizTotal TO 0, quizAvg TO 0.0;
    DISPLAY a message for user name
    WAIT for user to INPUT name

    FOR i = 0 to < 3
      DISPLAY a message asking the user to input number of items of Quiz #
      WAIT for user to INPUT quizScores
      DISPLAY a message asking the user to input score of Quiz #
      WAIT for user to INPUT quizScores

      ADD quizScores TO quizTotal
```

```
        DIVIDE quizTotal BY i TO GET quizAvg
    END FOR

    DISPLAY a message showing the results, student name

    FOR i = 0 to < 3
        DISPLAY quiz number, quizScores, quizTotal
        IF quizGrade >= 75.0
            DISPLAY a message showing the quiz is PASSED
        ELSE
            DISPLAY a message showing the QUIZ is FAILED
    ENDFOR

    DISPLAY the total score and average
    DISPLAY a message asking the user if they want to repeat
    WAIT for user input for y or n
  WHILE yesNo != 'n'

  DISPLAY Goodbye
END
```

Figure 2. Pseudocode of the Solution to Problem 1.

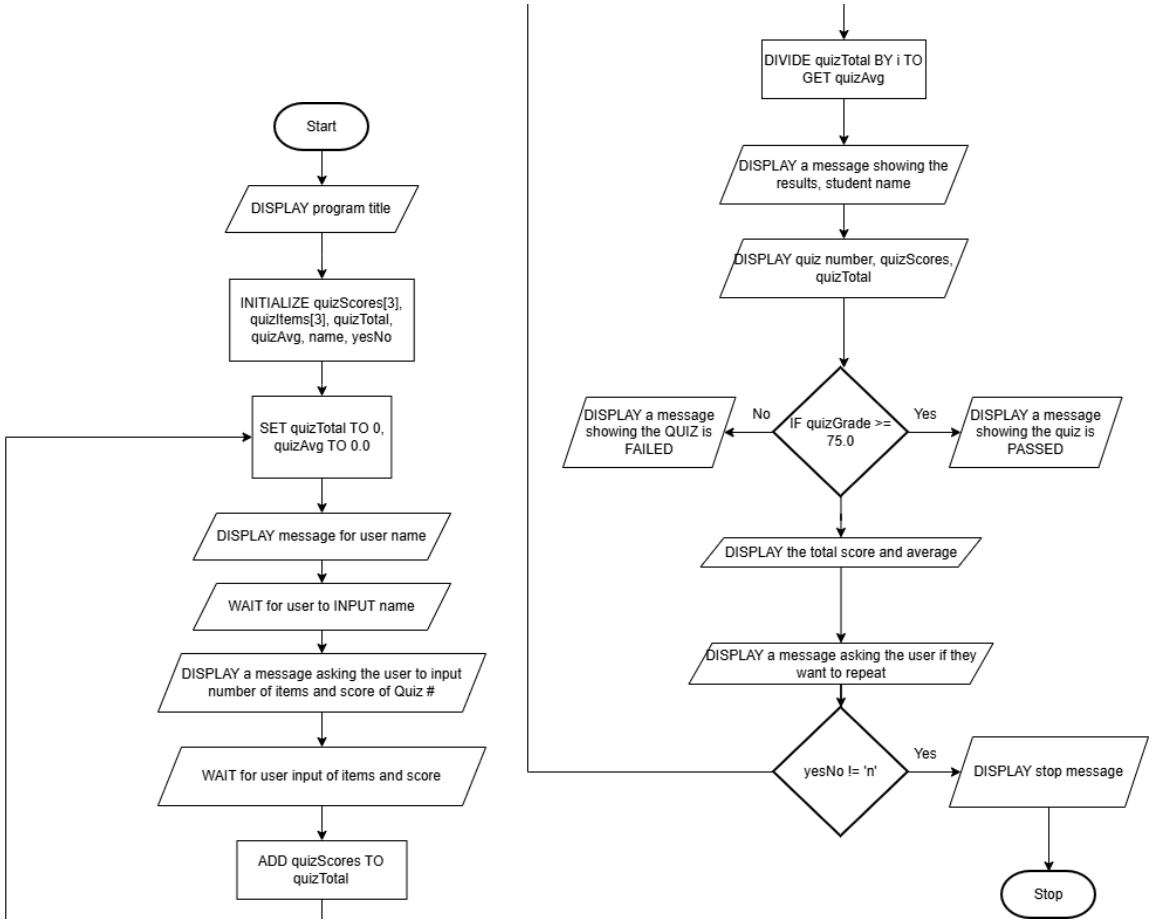


Figure 3. Flowchart of the Solution to Problem 1.

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main(){
    cout << "Seatwork 1.1 - Part 1\nComputing the Total & Average of 3 Quizzes\n";

    // Declare variables
    double quizScores[3], quizItems[3], quizTotal;
    float quizAvg;
    string name;
    char yesNo;

    do {
        // reset values
        quizTotal = 0; quizAvg = 0.0;

        // Prompt to input the student's name
        cout << "\nEnter the student's name: ";
        cin >> name;

        // Prompt to input the scores of 3 quizzes
        for (int i = 0; i < 3; i++){

            cout << "Enter the # of items of Quiz #" << (i + 1) << ": ";
            cin >> quizItems[i];
            cout << "Enter your score of Quiz #" << (i + 1) << ": ";
            cin >> quizScores[i];

            quizTotal += quizScores[i]; // Add scores to total
            quizAvg = quizTotal / (i + 1.0); // Compute average
        }

```

```
34
35          // Display the results
36          cout << "\nThe Results:\nStudent Name: " << name << "\nQuiz scores:";
37
38          // Display and process quiz grades
39          for (int j = 0; j < 3; j++){
40              cout << "\nQuiz #" << (j + 1) << ": " << quizScores[j] << "/" << quizItems[j];
41
42              float quizGrade = (quizScores[j] / quizItems[j]) * 50 + 50;
43              // if the score meets passing grade
44              if (quizGrade >= 75.0) {
45                  cout << " PASSED (" << fixed << setprecision(2) << quizGrade << "%)";
46              } else {
47                  cout << " FAILED (" << fixed << setprecision(2) << quizGrade << "%)";
48              }
49          }
50
51          cout << "\nThe total score of the 3 quizzes is: " << quizTotal;
52          cout << "\nThe average score of the 3 quizzes is: " << fixed << setprecision(2) << quizAvg;
53
54          // Prompt user to repeat
55          cout << "\n\nDo you want to repeat the program? (y/n): ";
56          cin >> yesNo;
57
58      } while (yesNo != 'n');
59
60      // when the user chooses not to repeat
61      cout << "\nGoodbye!\n";
62      return 0;
63  }
```

Figure 4. Code of the Solution to Problem 1.

This piece of code uses an array to store three numbers and a for loop system to do the process of adding numbers to the array 3 times. The program also includes a check to determine if the grade of the quiz is passing or failing. To allow the user to keep using the program, a do while loop is applied to the majority of the code's logic. Another for loop is used to display and process the quiz grades by reading the three pieces of data in the array. .

```
Seatwork 1.1 - Part 1
Computing the Total & Average of 3 Quizzes

Enter the student's name: Lara
Enter the # of items of Quiz #1: 20
Enter your score of Quiz #1: 15
Enter the # of items of Quiz #2: 25
Enter your score of Quiz #2: 13
Enter the # of items of Quiz #3: 30
Enter your score of Quiz #3: 15

The Results:
Student Name: Lara
Quiz scores:
Quiz #1: 15/20 PASSED (87.50%)
Quiz #2: 13.00/25.00 PASSED (76.00%)
Quiz #3: 15.00/30.00 PASSED (75.00%)
The total score of the 3 quizzes is: 43.00
The average score of the 3 quizzes is: 14.33

Do you want to repeat the program? (y/n): n

Goodbye!
```

Figure 5. Output of the Solution to Problem 1.

2. Cashiering system that computes for total price of the 3 items bought, accept payment and computes change. Display the transactions made.

1. Display program title
2. Do the following 3 times:
    a. Prompt the user to input the prices of 3 items
    b. Add the prices to the total
3. Display the total price
4. Prompt the user to input the payment. Repeat if payment is less than the total price
5. Compute the change
6. DIsplay the summary
7. Prompt the user to repeat the program. If not, the program ends.

Figure 1. Algorithm of the Solution to Problem 2.

```
START
  DISPLAY program title
  INITIALIZE itemPrices[3], totalPrice, payment, change, yesNo

  DO
    SET totalPrice to 0.0
```

```
    FOR i = 0 to < 3
       DISPLAY a message asking the user to input the prices
       WAIT for user to INPUT the prices
       ADD itemPrices TO totalPrice
    END FOR

    DISPLAY a message showing the total price
    DISPLAY a message asking the user to input the payment
    WAIT for user to input the payment
    DISPLAY a message showing the change amount

    DISPLAY summary of transactions

    DISPLAY a message asking the user if they want to repeat
    WAIT for user input for y or n

  WHILE yesNo != 'n'

  DISPLAY Goodbye
END
```

Figure 2. Pseudocode of the Solution to Problem 2.

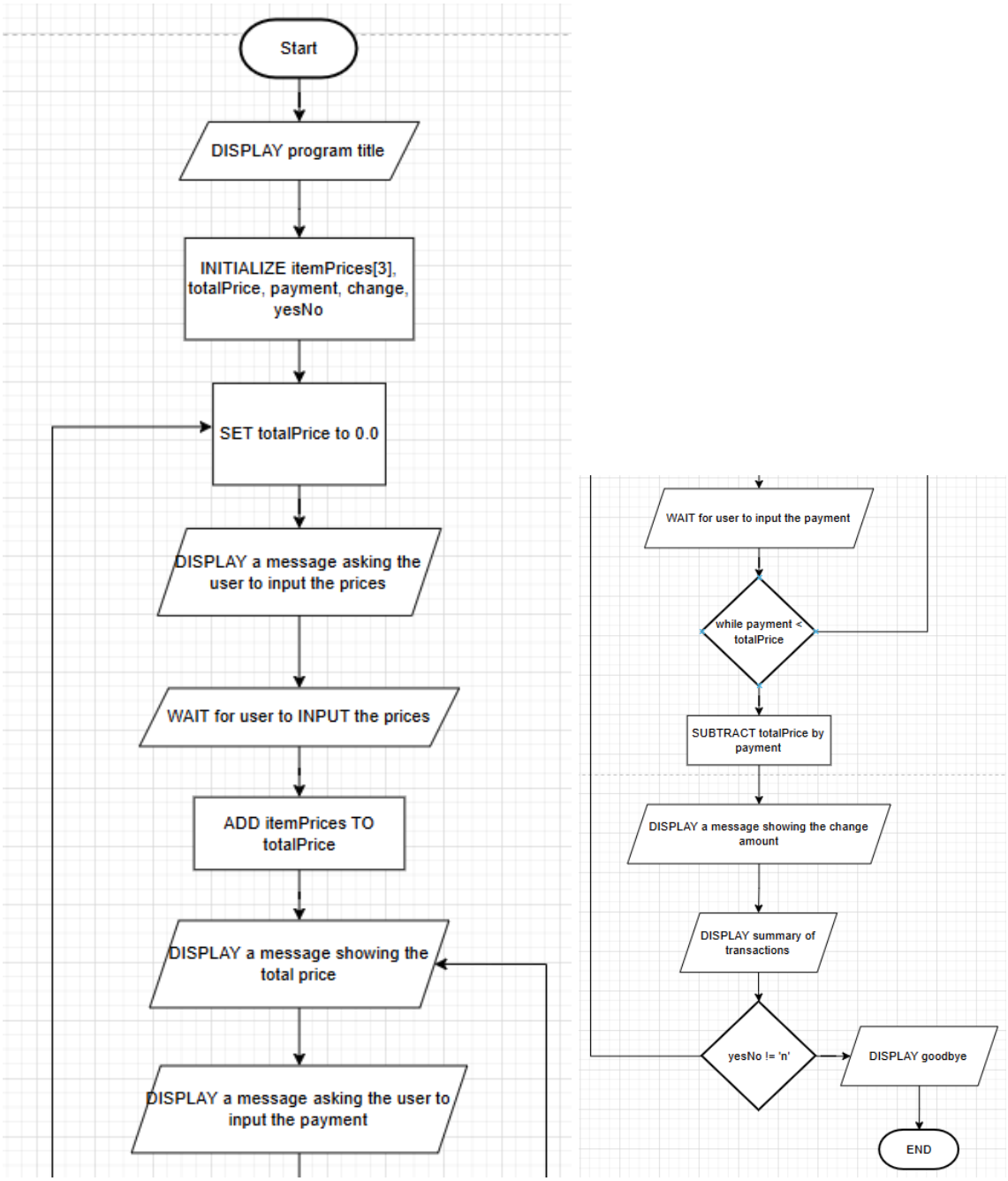


Figure 3. Flowchart of the Solution to Problem 2.

```cpp
#include <iostream>
#include <iomanip>

using namespace std;

int main(){
    cout << "Seatwork 1.1 - Part 2\nCashiering System\n\n";

    // Declare variables
    float itemPrices[3], totalPrice, payment, change;
    char yesNo;

    do {
        // reset values
        totalPrice = 0.0;

        // Prompt to input the prices of 3 items
        for (int i = 0; i < 3; i++){
        cout << "Enter the price of Item #" << (i + 1) << ": ";
        cin >> itemPrices[i];
        totalPrice += itemPrices[i]; // Add prices to total
        }

        // Compute the total price
        cout << "\nThe total price of the 3 items is: " << fixed << setprecision(2) << totalPrice;

        // Prompt to input the payment
        cout << "\nEnter the payment amount: ";
        cin >> payment;
        cout << "You entered: " << fixed << setprecision(2) << payment << endl;

        // Check if payment is sufficient, repeats if less than total price
        while (payment < totalPrice) {
            cout << "\nInsufficient payment amount. Please enter a valid payment amount: ";
            cin >> payment;
        }
        // Compute change
        change = payment - totalPrice;
        cout << "Your change is: " << fixed << setprecision(2) << change;

        cout << "\n\nTRANSACTIONS SUMMARY\nTotal Price: " << totalPrice << "\nAmount Paid: "
            << payment << "\nChange: " << change;

        // Prompt user to repeat
        cout << "\n\nDo you want to repeat the program? (y/n): ";
        cin >> yesNo;

    } while (yesNo != 'n');

    // when the user chooses not to repeat
    cout << "\nGoodbye!\n";
    return 0;
}
```

Figure 4. Code of the Solution to Problem 2

This code solution to Problem 2 contains arrays for storing data, for loops to read and write arrays, and a do-while loop to keep the program running if the user wishes to. A while check is present to check if the payment is sufficient to the total price of the three items bought. A summary of transactions is printed after the program's operations are done. To prevent prices from adding up, the necessary variables are reset to zero.

```
Seatwork 1.1 - Part 2
Cashiering System

Enter the price of Item #1: 24.99
Enter the price of Item #2: 31.99
Enter the price of Item #3: 99.99

The total price of the 3 items is: 156.97
Enter the payment amount: 150
You entered: 150.00

Insufficient payment amount. Please enter a valid payment amount: 160
Your change is: 3.03

TRANSACTIONS SUMMARY
Total Price: 156.97
Amount Paid: 160.00
Change: 3.03

Do you want to repeat the program? (y/n): n

Goodbye!
```

Figure 5. Output of the Solution to Problem 2.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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